

SQL Date Functions & Date Math

Q1 Day Name from Order Date

```
SELECT  
  order-id,  
  customer-id,  
  order-date,  
  DAYNAME (order-date) AS day-name  
FROM orders;
```

order-id	customer-id	order-date	day-name
1001	101	2026-05-01	Friday
1002	102	2026-05-02	Saturday
1003	103	2026-05-03	Sunday
1004	104	2026-05-04	Monday
1005	105	2026-05-05	Tuesday

Q2 Month Name from Signup Date

```
SELECT  
  customer-id,  
  customer-name,  
  signup-date,  
  MONTHNAME (signup-date) AS signup-month-name  
FROM customer-signups;
```

customer-id	customer-name	signup-date	signup-month-name
201	John	2026-01-15	January
202	Mary	2026-02-20	February
203	Peter	2026-03-05	March
204	Sarah	2026-04-18	April
205	Thabo	2026-05-30	May

Q3 Month Number from Sale Date

```

SELECT
    Sale-id,
    Product-name,
    Sale-date,
    MONTH (Sale-date) AS sale-month
FROM sales;

```

Sale-id	product-name	Sale-date	Sale-month
S001	Laptop	2026-01-10	1
S002	Mouse	2026-02-15	2
S003	keyboard	2026-03-20	3
S004	Monitor	2026-04-25	4
S005	Desk	2026-05-30	5

Q4 Year from Transaction Date

```

SELECT
    transacten-id,
    customer-id,
    transacten-date,
    YEAR (transacten-date) AS transaction-year
FROM transactens;

```

transacten-id	customer-id	transacten-date	transaction year
T001	101	2024-12-15	2024
T002	102	2025-01-20	2025
T003	103	2025-06-10	2025
T004	104	2026-02-05	2026
T005	105	2026-05-25	2026

Q5 Day of month from Delivery Date

```
SELECT  
  delivery-id,  
  customer-id,  
  delivery-date,  
  DAY (delivery-date) AS day-of-month  
FROM deliveries;
```

delivery-id	customer-id	delivery-date	day-of-month
D001	101	2026-05-01	1
D002	102	2026-05-08	8
D003	103	2026-05-15	15
D004	104	2026-05-22	22
D005	105	2026-05-29	29

Q6 Add Today's Date

```
SELECT  
  employee-id,  
  employee-name,  
  department,  
  CURRENT-DATE ( ) AS today-date  
FROM employees;
```

employee-id	employee-name	department	today-date
301	Nandi	Sales	2026-08-24
302	Brian	IT	2026-08-24
303	Lerato	Finance	2026-08-24
304	Sipho	HR	2026-08-24
305	Aisha	Marketing	2026-08-24

Q7 Convert Text to Date

```
SELECT
    order-id,
    customer-id,
    order-date-text,
    TO-DATE (order-date-text) AS order-date
FROM online-orders;
```

order-id	customer-id	order-date-text	order-date
4001	101	2026-01-15	2026-01-15
4002	102	2026-02-20	2026-02-20
4003	103	2026-03-25	2026-03-25
4004	104	2026-04-10	2026-04-10
4005	105	2026-05-05	2026-05-05

Q8 Format Date As Text

```
SELECT
    payment-id,
    customer-id,
    payment-date,
    TO CHAR (payment-date, 'YYYY-MM-DD') AS formatted-payment-date
FROM payment-dates;
```

payment-id	customer-id	payment-date	formatted-payment-date
P001	101	2026-01-05	2026-01-05
P002	102	2026-02-10	2026-02-10
P003	103	2026-03-15	2026-03-15
P004	104	2026-04-20	2026-04-20
P005	105	2026-05-25	2026-05-25

Q9 Days Since Last Purchase

SELECT

customer-id,

customer-name,

last-purchase-date;

DATEDIFF(CURRENT-DATE(), last-purchase-date) AS days-since-last-purchase

FROM customer-purchases;

customer-id	customer-name	last-purchase-date	days-since-last-purchase
501	John	2026-05-01	115
502	Mary	2026-05-10	106
503	Peter	2026-05-15	101
504	Sarah	2026-05-20	96
505	Thabo	2026-05-25	91

Q10 Expected Delivery Date

SELECT

order-id,

customer-id,

order-date,

DATEADD(DAY, 7, order-date) AS expected-delivery-date

FROM shipping-orders;

order-id	customer-id	order-date	expected-delivery-date
6001	101	2026-05-01	2026-05-08
6002	102	2026-05-03	2026-05-10
6003	103	2026-05-05	2026-05-12
6004	104	2026-05-07	2026-05-14
6005	105	2026-05-09	2026-05-16

Q11 Extract Year, Month and Day

```
SELECT
    booking-id,
    customer-id,
    booking-date,
    YEAR (booking-date) AS booking-year,
    MONTH (booking-date) AS booking-month,
    DAY (booking-date) AS booking-day,
    FROM bookings;
```

booking-id	customer-id	booking-date	booking-year	booking-month	booking-day
B001	101	2026-01-12	2026	1	12
B002	102	2026-02-18	2026	2	18
B003	103	2026-03-22	2026	3	22
B004	104	2026-04-09	2026	4	9
B005	105	2026-05-27	2026	5	27

Q12 Filter orders from 2026

```
SELECT
    order-id,
    customer-id,
    order-date,
    YEAR (order-date) AS order-year,
    amount
    FROM yearly-orders
    WHERE YEAR (order-date) = 2026;
```

order-id	customer-id	order-date	order-year	amount
7004	104	2026-02-05	2026	1500
7005	105	2026-05-25	2026	2000

Q13 Filter orders from March

```

SELECT
    order-id,
    customer-id,
    order-date,
    MONTH (order-date) AS order-month,
    amount
FROM monthly-orders
WHERE MONTH (order-date) = 3;
    
```

order-id	customer-id	order-date	order-month	amount
8003	103	2026-03-10	3	8000
8004	104	2026-03-25	3	1500

Q14 Last Day of the Month

```

SELECT
    subscription-id,
    customer-id,
    start-date,
    LAST-DAY (start-date) AS month-end-date
FROM subscriptions;
    
```

Subscription-id	customer-id	start-date	month-end-date
SUB 001	101	2026-01-10	2026-01-31
SUB 002	102	2026-02-15	2026-02-28
SUB 003	103	2026-03-20	2026-03-31
SUB 004	104	2026-04-25	2026-04-30
SUB 005	105	2026-05-30	2026-05-31

Q15 First day of the Month

```

SELECT
    send-id,
    customer-id,
    send-date,
    DATE-TRUNC ('month', send-date) AS month-start-date
FROM campaign-sends;

```

send-id	customer-id	send-date	month-start-date
C001	101	2026-01-12	2026-01-01
C002	102	2026-02-18	2026-02-01
C003	103	2026-03-22	2026-03-01
C004	104	2026-04-09	2026-04-01
C005	105	2026-05-27	2026-05-01

Q16 Format as Month and Year

```

SELECT
    invoice-id,
    customer-id,
    invoice-date,
    TO-CHAR (invoice-date, 'Month YYYY') AS invoice-month-year
FROM invoice-dates;

```

invoice-id	customer-id	invoice-date	invoice-month-year
INV 001	101	2026-01-05	January 2026
INV 002	102	2026-02-10	February 2026
INV 003	103	2026-03-15	March 2026
INV 004	104	2026-04-20	April 2026
INV 005	105	2026-05-25	May 2026

Q17 Calculate Customer Age from Date of Birth

SELECT

customer-id,
customer-name,
date-of-birth,

YEAR(CURRENT-DATE()) - YEAR(date-of-birth)

CASE

WHEN DATE-FORMAT(CURRENT-DATE(), '%m-%d')
< DATE-FORMAT(date-of-birth, '%m-%d')

THEN 1

ELSE 0

END AS customer-age

FROM customer-birthdays;

customer-id	customer-name	date-of-birth	customer-age
901	John	1998-05-10	28
902	Mary	1990-08-20	36
903	Peter	2002-03-15	24
904	Sarah	1985-12-01	40
905	Thebo	2000-07-30	26

Q18 Classify Weekend vs Weekday

SELECT

order-id,
customer-id,
order-date,

DAYNAME(order-date) AS day-name,

CASE

WHEN DAYNAME(order-date) IN ('Saturday', 'Sunday')

THEN 'weekend'

ELSE 'weekday'

END AS day-type

FROM weekend-orders;

order-id	customer-id	order-date	day-name	day-type
9001	101	2026-05-01	Friday	weekday
9002	102	2026-05-02	Saturday	weekend
9003	103	2026-05-03	Sunday	weekend
9004	104	2026-05-04	Monday	weekday
9005	105	2026-05-05	Tuesday	weekday

Q19 Transaction by Quarter

```

SELECT
    transaction-id,
    customer-id,
    transaction-date,
    QUARTER (transaction-date) AS transaction-quarter,
    amount
FROM quarterly-transactions;
    
```

transaction-id	customer-id	transaction-date	transaction-quarter	amount
Q 001	101	2026-01-15	1	500
Q 002	102	2026-03-20	1	1200
Q 003	103	2026-04-10	2	800
Q 004	104	2026-07-05	3	1500
Q 005	105	2026-10-25	4	2000

Q20 orders Older than 30 Days

```

SELECT
    order-id,
    customer-id,
    order-date,
    DATEDIFF (day, order-date, CURRENT-DATE ()) AS days-since-order,
    amount
FROM recent-orders
WHERE DATEDIFF (day, order-date, CURRENT-DATE ()) > 30;
    
```

order-id	customer-id	order-date	day-since-order	amount
R001	101	2026-04-01	146	500
R002	102	2026-04-15	132	1200
R003	103	2026-05-01	116	800
R004	104	2026-05-10	107	1500
R005	105	2026-05-25	92	2000

BONUS CHALLENGE

customer Recency Segment

SELECT

customer - id ,

customer - name ,

last - purchase - date ,

DATE DIFF (day , last - purchase - date , CURRENT - DATE ())

AS days - since - last - purchase

CASE

WHEN DATE DIFF (day , last - purchase - date , CURRENT - DATE ()) $\leq$ 30
THEN 'Active Customer'

WHEN DATE DIFF (day , purchase - date , CURRENT - DATE ()) BETWEEN 31 AND 90
THEN 'At Risk Customer'

ELSE 'Inactive Customer'

END AS customer - status

FROM customer - recency ;

customer - id	customer - name	last - purchase - date	days - since - last - purchase	customer - status
101	John	2026-05-25	92	Inactive Customer
102	Many	2026-05-10	107	Inactive Customer
103	Peter	2026-04-01	146	Inactive Customer
104	Sarah	2026-02-15	191	Inactive Customer
105	Thabo	2026-12-20	248	Inactive Customer